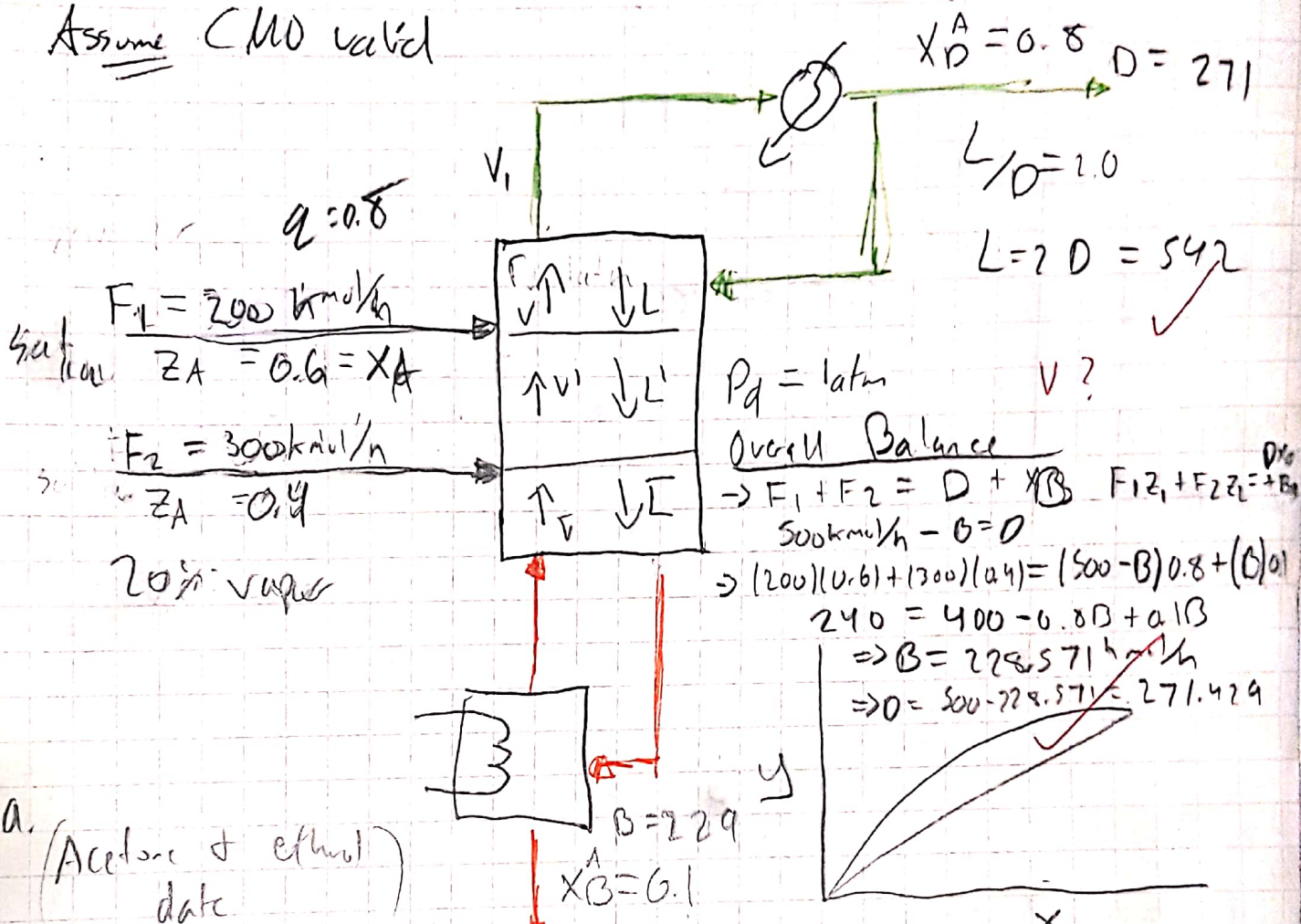


Q9: Separating Acetone + ethanol

9

• Total condenser & partial reboiler

Assume CMO valid



a. (Acetone + ethanol)
data

• 2 Feed lines, 1 top OP, 1 Bottom OP, 1 middle OP

Feed lines: $y = \frac{q}{q-1}x - \frac{z_i}{q-1}$

Feed 1: Sat Liq $\Rightarrow q = \frac{L}{F} = 1$

$y = \frac{1}{1-1}x - \frac{z_i}{1-1} \Rightarrow x_A = z_A$ (vertical line)

Feed 2: 20% vapor
80% L
 $q = \frac{L}{F} = 0.8$

$z_A = 0.4$

$y = \frac{0.8}{0.8-1}x - \frac{z_i}{0.8-1} = -4x + 5z_i = -4x + 2$

TOP OP

$$y = \frac{L}{v}x + (1 - \frac{L}{v})x_0 \quad \checkmark$$

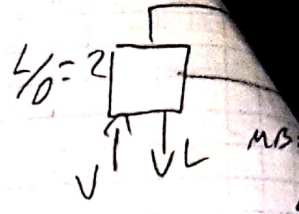
$$y = \frac{L}{L+D}x + (1 - \frac{L}{L+D})x_0$$

$$y = \frac{L/0}{L/0+0/0}x + (1 - \frac{L/0}{L/0+0/0})x_0$$

$$y = \frac{0.2}{2+1}x + (1 - \frac{0.2}{2+1})x_0$$

$$y = \frac{2}{3}x + \frac{1}{3}x_0$$

$$x_0 = 0.8$$



Bottom OP

$$y = \frac{L}{v}x - (\frac{L}{v} - 1)x_B$$

Middle OP

$$y = \frac{L'}{v}x + \frac{D \times 0 - F_2 Z_1}{v}$$

$$L' = F_1' Z_1 + L$$

$$= (300)(1) + 542$$

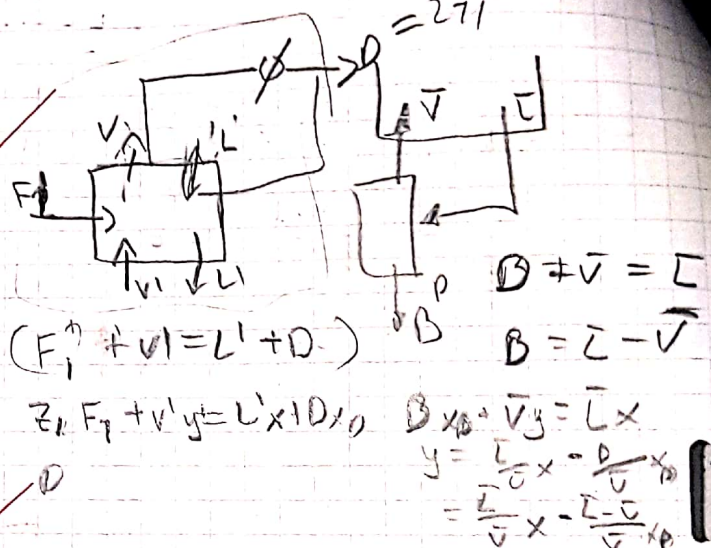
$$(L' = 742 \text{ kmol/h})$$

$$v' = L' + D - F_1$$

$$(v' = 742 + 271 - 200 = 813)$$

$$y = \frac{742}{813}x + \frac{(0.2)(271) - 200(0.61)}{813}$$

$$(y = 0.913x + 0.1145)$$



$$(F_1' + v' = L' + D)$$

$$Z_1 F_1 + v' y = L' x + D x_0$$

$$B x_0 + \bar{v} y = \bar{L} x$$

$$y = \frac{L}{v}x - \frac{D}{v}x_0$$

$$= \frac{L}{v}x - \frac{L-D}{v}x_0$$

Bottom OP

$$L' + F_2 q = \bar{L}$$

$$\bar{L} = 742 + 300 \cdot 0.9 = 982 \text{ kmol/h}$$

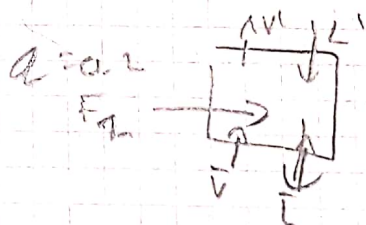
$$\bar{v} + F_2 (1-q) = v'$$

$$(\bar{v} = v' - F_2 (1-q) = 813 - 300(1-0.9) = 753 \text{ kmol/h})$$

$$\Rightarrow y = \frac{982}{753}x - (\frac{982}{753} - 1)0.1$$

$$(y = 1.3041x - 0.10309)$$

$$\left[\frac{\bar{v}}{B} = \frac{753}{228.51} = 3.29 \right] \checkmark$$



⇒ Drawing lines in graph from x_D to $y=x$
 ⇒ 6 stages ✓

⇒ From graph: Feed 1 → stage 2
 where feed line intercepts "box"
 Feed 2 → stage 4 ✓

(b) $(L/D)_{min} = \frac{x_D - y^*}{x_D - x^*}$

$y^* = 0.74$ $x^* = 0.6$ $x_D = 0.8$

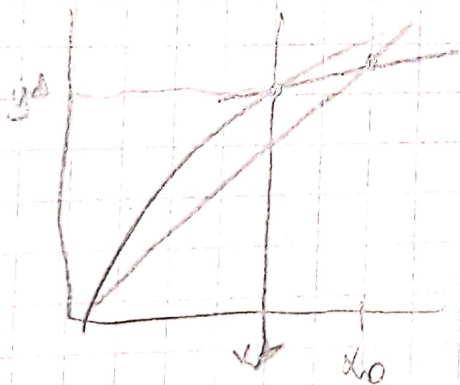
$(L/D)_{min} = \frac{0.8 - 0.74}{0.8 - 0.6} = 0.3$ ✓

$(L/D)_{min} = \left(\frac{L}{V-L} \right)_{min}^{\frac{V}{V-L}} = \left(\frac{\frac{L}{V}}{\frac{V-L}{V-L}} \right)_{min}$

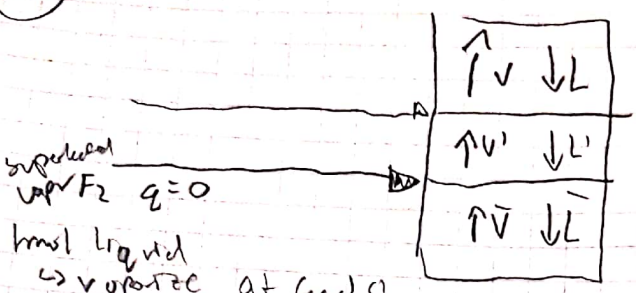
$\left(\frac{L}{D} \right)_{min} = \frac{(L/V)_{min}}{1 - (L/V)_{min}}$

$\left(\frac{L}{D} \right)_{min} = \frac{0.3}{1 - 0.3}$

$\left(\frac{L}{D} \right)_{min} = 0.429$ ✓



(C.)



5 moles feed
1 mol liquid



$$h_{total} = ?$$

$$D = ?$$

$$\left[\frac{h_{total}}{D} = 4 \right]$$

$$h_{total} = h_L + h_{HL} + h_F$$

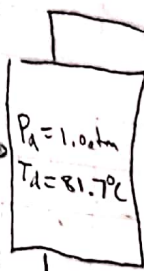
DIC Methanol-water

$$F = 1000 \text{ lbmol/h}$$

$$z_m = 0.3$$

$$z_w = 0.7$$

$$\sum x \text{ or } y \text{ or } z = 1$$



$$y \rightarrow \frac{\text{height}}{\text{stage}} + \text{temp}$$

$$V = 264 \text{ lbmol/h}$$

$$\frac{V}{F} = f = 0.264$$

$$MW_{mix} = (x MW)_m + (x MW)_w$$

$$x = 0.2$$

$$L = 736 \text{ lbmol/h}$$

MB: ①

$$F = V + L$$

$$\Rightarrow L = F - V$$

$$Fz = Vy + Lx$$

$$Vy = -Lx + Fz$$

$$y = -\frac{L}{V}x + \frac{F}{V}z$$

$$y = \left(-\frac{F-V}{V} \right) x + \frac{F}{V}z$$

$$= -\frac{F-F+V}{V} x + \frac{F}{V}z$$

$$(y = -\frac{1-f}{f}x + \frac{f}{f}z)$$

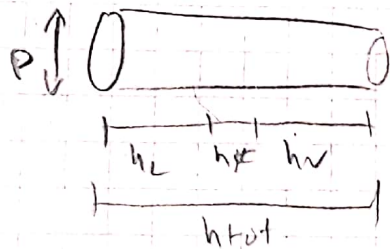
$$fy = (f-1)x + z$$

$$fy - fx = -x + z$$

$$f(y-x) = -x + z$$

$$f = \frac{-x + z}{y-x} = \frac{-0.2 + 0.3}{0.574 - 0.2} = 0.264$$

$$E_{ans} \mu_{pen} = h_{drum} \sqrt{\frac{\rho_L - \rho_V}{\rho}}$$



units

① Liquid density

$$[MW_L = x_m MW_m + x_w MW_w]$$

$$= (0.2)(32.043) + (0.8)(18.016)$$

$$MW_m = 32.043 \text{ lb/lbmol}$$

$$MW_w = 18.016$$

$$MW_L = 20.824 \text{ lb/lbmol}$$

$$[\bar{V}_L = x_m \bar{V}_m + x_w \bar{V}_w = x_m \frac{MW_m}{\rho_m} + x_w \frac{MW_w}{\rho_w}]$$

$$= 0.2 \left(\frac{32.043 \text{ lb/lbmol}}{49.85 \text{ lb/ft}^3} \right) + 0.8 \left(\frac{18.016 \text{ lb/lbmol}}{62.4 \text{ lb/ft}^3} \right)$$

$$\bar{V}_L = 0.360 \text{ ft}^3/\text{lbmol}$$

$$[\rho_L = \frac{MW_L}{\bar{V}_L}] = \frac{20.824 \text{ lb/lbmol}}{0.360 \text{ ft}^3/\text{lbmol}} = 57.768 \text{ lb/ft}^3$$

D14 / ..

② Vapor Density

$$\begin{aligned} \overline{M}_{wv} &= y_m \overline{M}_{wm} + y_w \overline{M}_{ww} \\ &= (0.579)(32.045 \frac{\text{lb}}{\text{lbmol}}) + (0.421)(18.016 \frac{\text{lb}}{\text{lbmol}}) \\ \overline{M}_{wv} &= 26.136 \frac{\text{lb}}{\text{lbmol}} \end{aligned}$$

$$PV = nRT$$

$$\frac{n}{V} = \hat{\rho}_v = \frac{P}{RT}$$

$$\rho_v = \overline{M}_{wv} \hat{\rho}_v = \frac{(1.0 \text{ atm})(26.136 \frac{\text{lb}}{\text{lbmol}})}{(0.73029 \frac{\text{ft}^3}{\text{lbmol} \cdot \text{R}})(51.70 \text{ R})} = \frac{635.72 \text{ lb}}{37.769 \text{ ft}^3}$$

$$\rho_v = 0.0554 \frac{\text{lb}}{\text{ft}^3}$$

$$0.8999 \frac{\text{lb}}{\text{ft}^3}$$

③ k_{drum} Calculation

$$k_{drum} = (\text{constant}) \exp(A + B \ln F_{iv} + C(\ln F_{iv})^2 + D(\ln F_{iv})^3 + E(\ln F_{iv})^4)$$

$$F_{iv} = \frac{W_L}{W_v} \sqrt{\frac{\rho_v}{\rho_L}} \quad \text{or } < 0.605$$

$$\Rightarrow W_v = (V) \overline{M}_{wv} = (264 \frac{\text{lbmol}}{\text{h}})(26.136 \frac{\text{lb}}{\text{lbmol}}) = 6,898.9 \frac{\text{lb}}{\text{h}}$$

$$\rightarrow W_L = (L) \overline{M}_{wL} = (736 \frac{\text{lbmol}}{\text{h}})(20.824 \frac{\text{lb}}{\text{lbmol}}) = 15,326 \frac{\text{lb}}{\text{h}}$$

$$\Rightarrow F_{iv} = \frac{W_L}{W_v} \sqrt{\frac{\rho_v}{\rho_L}} = \frac{15,326}{6,898.9} \sqrt{\frac{0.0554}{37.769}} = 0.0658$$

$$\text{calc } k_{drum} = 0.444 \frac{\text{ft}}{\text{s}}$$

$$\text{④ } \mu_{perm} = k_{drum} \sqrt{\frac{\rho_L - \rho_v}{\rho_v}} = (0.444 \frac{\text{ft}}{\text{s}}) \sqrt{\frac{37.769 - 0.0554}{0.0554}} = 17.938 \frac{\text{ft}}{\text{s}}$$

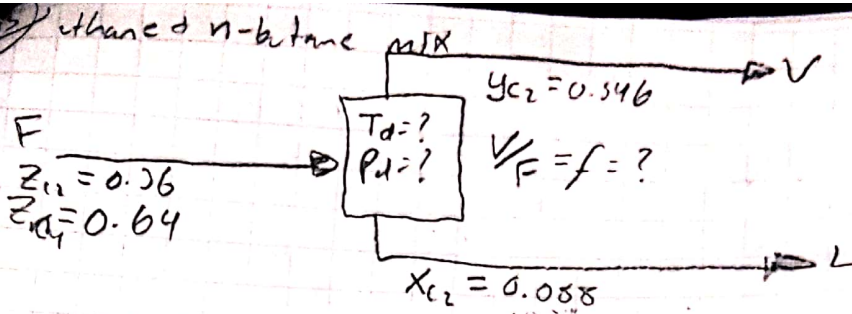
④ Area & Diameter

$$A_c = \frac{V(\overline{M}_{wv})}{\mu_{perm} (3600) \rho_v} = \frac{(264 \frac{\text{lbmol}}{\text{h}})(26.136 \frac{\text{lb}}{\text{lbmol}})}{(17.938 \frac{\text{ft}}{\text{s}})(3600)(0.0554 \frac{\text{lb}}{\text{ft}^3})} = 7.929 \text{ ft}^2$$

$$A_{tot} = \frac{A_c}{0.7} = 9.643 \text{ ft}^2$$

$$D = \sqrt{\frac{4A_c}{\pi}} = \sqrt{\frac{4(9.643)}{\pi}} = 3.504 \text{ ft}$$

$$\frac{h_{total}}{D} = 4 \Rightarrow h_{total} = 4D = 14.017 \text{ ft}$$



$$\sum x = 1$$

$$x_{14} = 1 - x_{12}$$

$$= 1 - 0.088$$

$$x_{14} = 0.912$$

$$\sum y = 1$$

$$y_{14} = 1 - y_{12}$$

$$= 1 - 0.546$$

$$y_{14} = 0.454$$

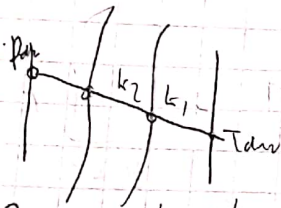
$$B: F = V + L \Rightarrow L = F - V$$

$$Fz = Vy + Lx$$

$$y = -\frac{L}{V}x + \frac{F}{V}z$$

$$y = -\frac{1-f}{f}x + \frac{1}{f}z$$

$$f = \frac{-x + z}{y - x} = \frac{-0.088 + 0.36}{0.546 - 0.36} = 0.533$$



$$k_{12} = \frac{y_{12}}{x_{12}} = \frac{0.546}{0.088} = 6.205$$

$$k_{14} = \frac{y_{14}}{x_{14}} = \frac{0.454}{0.912} = 0.498$$

De protractor $\Rightarrow$ line through k_1 + k_2

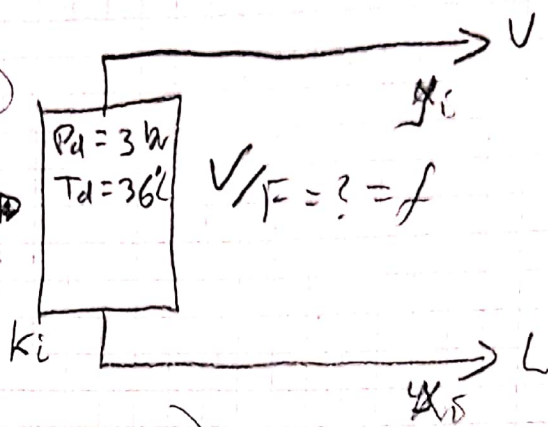
$$P_d = 360 \text{ kPa}$$

$$T_d = 16^\circ\text{C}$$

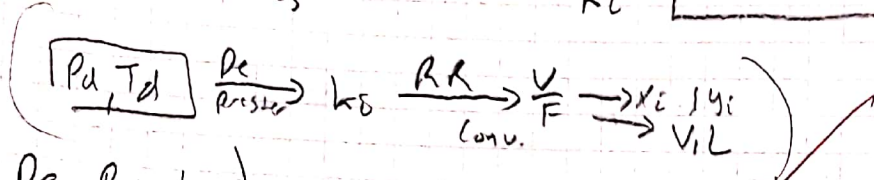
D14) Mix; (1, 1, 4, 1, 5)

$P_F = 5 \text{ bar}$
 $T_F = 50.40^\circ\text{C}$
 $F = 172 \text{ kmol/h}$

$V + L$



$Z_{c1} = 0.12$
 $Z_{c4} = 0.48$
 $Z_{c5} = 0.4$



P_F Priester: $T_d + P_d \Rightarrow k_{c1} = 50$ $k_{c4} = 1.1$ $k_{c5} = 0.35$

$$f(V/F) = -\sum_{i=1}^3 \frac{(k_i - 1)Z_i}{1 + (k_i - 1)V/F} = \frac{(k_{c1} - 1)Z_{c1}}{1 + (k_{c1} - 1)V/F} + \frac{(k_{c4} - 1)Z_{c4}}{1 + (k_{c4} - 1)V/F} + \frac{(k_{c5} - 1)Z_{c5}}{1 + (k_{c5} - 1)V/F}$$

(1) $f(V/F) = \frac{5.88}{1 + 49V/F} + \frac{0.018}{1 + 0.1V/F} + \frac{-0.26}{1 - 0.65V/F}$

Newtonian Conv.

$$\frac{df}{d(V/F)} = -\sum_{i=1}^3 \frac{(k_i - 1)Z_i}{[1 + (k_i - 1)V/F]^2}$$

$$= -\left[\frac{288.12}{(1 + 49V/F)^2} + \frac{0.0048}{(1 + 0.1V/F)^2} + \frac{0.169}{(1 - 0.65V/F)^2} \right]$$

Trial + error $V/F = 0.1$ Guess

$\frac{df}{d(V/F)}|_{0.1} = -8.475 \rightarrow f(0.1) = 0.766$

2nd: $(V/F)_2 = (V/F)_1 - \frac{f_1}{df/d(V/F)} = 0.190$

$\rightarrow \frac{df}{d(V/F)}|_2 = -2.925 \rightarrow f(0.190) = 0.3196$

3rd: $(V/F)_3 = (V/F)_2 - \frac{f_2}{df/d(V/F)} = 0.29966$

$\rightarrow \frac{df}{d(V/F)}|_3 = -1.4365 \rightarrow f(0.29966) = 0.0986$

4th: $(V/F)_4 = (V/F)_3 - \frac{f_3}{df/d(V/F)} = 0.36831$

$\rightarrow \frac{df}{d(V/F)}|_4 = -1.0407 \rightarrow f(0.36831) = 0.01315$

5th: $(V/F)_5 = (V/F)_4 - \frac{f_4}{df/d(V/F)} = 0.38038$

$\rightarrow \frac{df}{d(V/F)}|_5 = -1.0197 \rightarrow f(0.38038) = 2.516 \times 10^{-4} < 0.005$

$$V/F = 0.38038$$

$$V = 0.38038 F$$

$$V = 46.406 \text{ kmol/h}$$

$$X_i = \frac{Z_i}{1 + V_F (K_i - 1)}$$

$$X_{C1} = \frac{0.12}{1 + (50 - 1) \cdot 0.38038} = 0.0061$$

$$X_{C4} = \frac{0.48}{1 + (1.1 - 1) \cdot 0.38038} = 0.4624$$

$$X_{C5} = \frac{0.4}{1 + (0.35 - 1) \cdot 0.38038} = 0.53138$$

$$L = 7 - V$$

$$= 122 - 46.406$$

$$L = 75.5935 \text{ kmol/h}$$

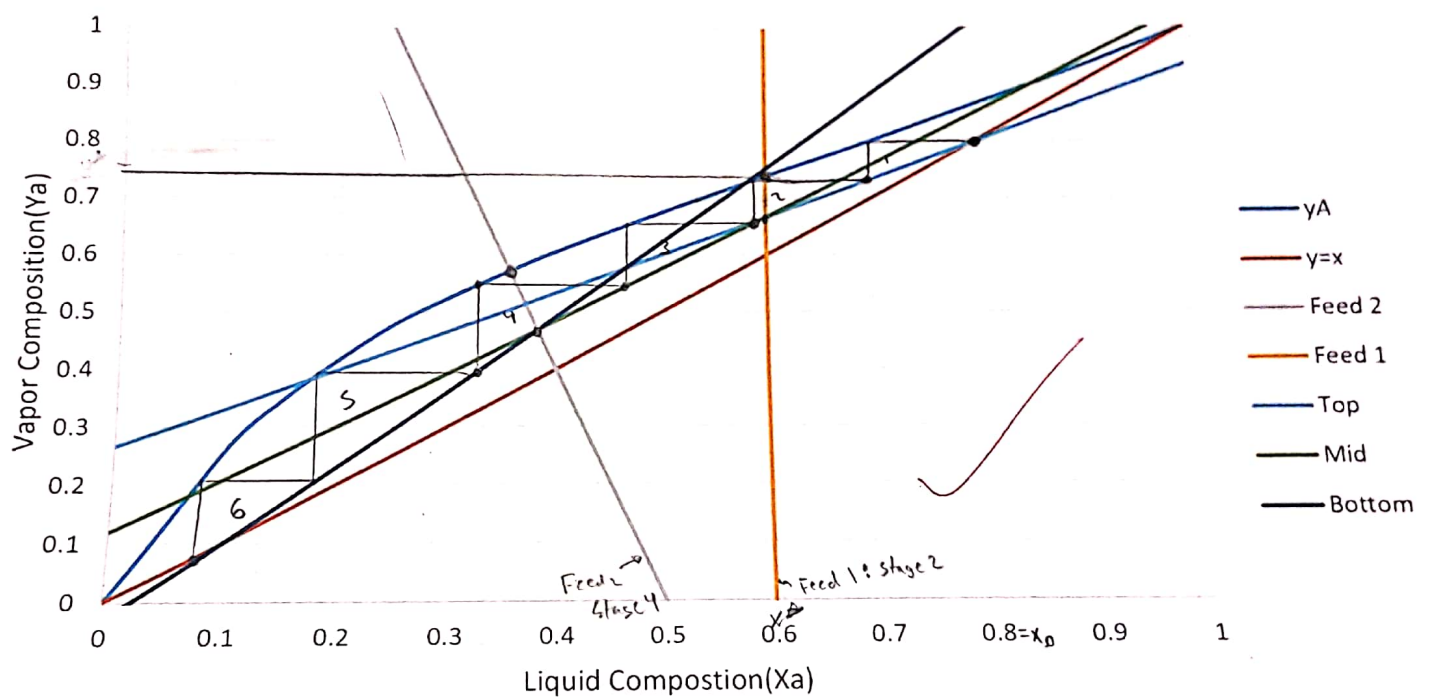
$$y_i = \frac{R_i X_i}{L}$$

$$y_{C1} = (50 \times 0.0061) = 0.3055$$

$$y_{C4} = (1.1 \times 0.4624) = 0.5086$$

$$y_{C5} = (0.35 \times 0.53138) = 0.18598$$

McCabe-Thiele Diagram for Acetone in Ethanol



xA	yA	x	y = x	x	Feed 1	x	Feed 2	x	Top	x	Bottom	x	Mid
0	0	0	0	0.6	0	0	2	0	0.266	0	-0.0304	0	0.119
0.1	0.262	0.05	0.05	0.6	0.1	0.05	1.8	0.05	0.299333	0.05	0.034805	0.05	0.16465
0.15	0.348	0.1	0.1	0.6	0.2	0.1	1.6	0.1	0.332667	0.1	0.10001	0.1	0.2103
0.2	0.417	0.15	0.15	0.6	0.3	0.15	1.4	0.15	0.366	0.15	0.165215	0.15	0.25595
0.25	0.478	0.2	0.2	0.6	0.4	0.2	1.2	0.2	0.399333	0.2	0.23042	0.2	0.3016
0.3	0.524	0.25	0.25	0.6	0.5	0.25	1	0.25	0.432667	0.25	0.295625	0.25	0.34725
0.35	0.566	0.3	0.3	0.6	0.6	0.3	0.8	0.3	0.466	0.3	0.36083	0.3	0.3929
0.4	0.605	0.35	0.35	0.6	0.7	0.35	0.6	0.35	0.499333	0.35	0.426035	0.35	0.43855
0.5	0.674	0.4	0.4	0.6	0.8	0.4	0.4	0.4	0.532667	0.4	0.49124	0.4	0.4842
0.6	0.739	0.45	0.45	0.6	0.9	0.45	0.2	0.45	0.566	0.45	0.556445	0.45	0.52985
0.7	0.8	0.5	0.5	0.6	1	0.5	0	0.5	0.599333	0.5	0.62165	0.5	0.5755
0.8	0.865	0.55	0.55					0.55	0.632667	0.55	0.686855	0.55	0.62115
0.9	0.929	0.6	0.6					0.6	0.666	0.6	0.75206	0.6	0.6668
1	1	0.65	0.65					0.65	0.699333	0.65	0.817265	0.65	0.71245
		0.7	0.7					0.7	0.732667	0.7	0.88247	0.7	0.7581
		0.75	0.75					0.75	0.766	0.75	0.947675	0.75	0.80375
		0.8	0.8					0.8	0.799333	0.8	1.01288	0.8	0.8494
		0.85	0.85					0.85	0.832667	0.85	1.078085	0.85	0.89505
		0.9	0.9					0.9	0.866	0.9	1.14329	0.9	0.9407
		0.95	0.95					0.95	0.899333	0.95	1.208495	0.95	0.98635
		1	1					1	0.932667	1	1.2737	1	1.032